Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/26/2005
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Pool Boiling Heat Transfer Mechanisms in Microgravity and Partial Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	12/22/2003	**End Date:**	10/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2007 per PI (11/06). Note: extended 4 more years		
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	NNC04AA26A		
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Task Description:	The objective of the proposed research is to determine the effects of microgravity on nucleate boiling heat transfer. An understanding of microgravity effects on boiling mechanisms is critical to the proper design of phase change heat removal equipment for use in space-based applications. Experiments to date have shown that stable, subcooled boiling on flat plates in microgravity environments is possible. It is presently thought that coalescence and condensation of bubbles away from the heater cause oscillatory motions within the surrounding liquid, thereby providing a continual supply of liquid from the bulk of the pool to the base of the bubble just above the heater. Almost all research pertaining to boiling in microgravity environments, however, has thus far been either of a qualitative nature (photographic studies) with some average wall heat flux/wall temperature measurements, analytical work, or numerical simulations. In the studies where the heat transfer coefficients were measured, the heated surface was always comparable to or much larger than the bubble sizes, so only average heat transfer rates over the entire heated surface were obtained. Very little experimental data is available regarding the local heat transfer rates under and around the bubbles as they grow and depart from the surface. Such information, especially when correlated with visual studies, can provide much needed information regarding the relevant wall heat transfer mechanisms during the bubble departure cycle by pinpointing when and where in the cycle large amounts of heat are removed. Microgravity effects on critical heat flux levels (the maximum heat flux level that can be attained without a catastrophic increase in heater temperature) is another very important area that must be addressed if boiling is to be used reliably as a heat removal mechanism. Although critical heat flux is known to decrease with decreasing gravity, very little quantitative information is available due to the use of constant wall heat flux heaters. A four-year project with two primary objectives is proposed: 1) to determine the relative contributions of various heat transfer mechanisms to the overall heat flux in subcooled, nucleate pool boiling on a flat plate in microgravity environments, and 2) to obtain quantitative data on the effects of microgravity on critical heat flux levels. The research will results in better understanding of the mechanisms of boiling heat transfer, and should enable the development of more efficient two-phase heat exchangers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research has shown that transient conduction is the dominant heat transfer mechanism in boiling of refrigerants-like fluids.
Task Progress:	1. Science Progress and Status Although thermocapillary convection in organics has been observed by numerous researchers, its origins are unknown. Recent research has suggested that small amounts of dissolved gas can accumulate in the bubbles, causing the vapor pressure (and therefore the saturation temperature and surface tension) to vary along the bubble surface. The strength of thermocapillary convection has been observed to alter the wall heat transfer by changing the size of the primary bubble, allowing additional liquid to wet the surface. For the past year, we have focused on finding an explanation for the Marangoni convection that was observed during the low-g flights. Possible explanations for thermocapillary convection (dissolved gas, fluid contaminants) were systematically tested, and additional experiments and numerical simulations are planned. We again used the microheater array held at constant temperature to measure the wall heat transfer. The heater array and the associated electronics were packaged in a test rig that could be flown on the KC-135. Boiling curves were obtained when heaters of various aspect ratio (2x2, 2x4, 2x6, 2x8, 3x3, 6x6, 10x10) were powered. Care was taken to thoroughly degas the liquid to levels below 3 ppm by repeatedly removing the vapor/air above the liquid and measuring the pressure and temperature of the liquid. Thermocapillary convection dominated boiling was observed at the highest superheat (34.7 °C). The size of the bubbles on all of the heaters decreased as the superheat increased, which we believe was due to increased thermocapillary flow causing enhanced condensation on the top of the bubbles. More of the heater surface was wetted by liquid, allowing nucleation to occur. Strong thermocapillary convection was observed even for gas concentrations in the liquid <3 ppm. CHF was not observed for the heater used in this study indicating that CHF in low-g may be highly dependent on the surface characteristics of the heater, heater size, and the dissolved gas concentration due primarily to their effect on nucleation site density. For large heaters (7x7 mm2) CHF may was completely dominated by thermocapillary convection. Also, the wall heat transfer was observed to be similar in low-g and high-g when there was very little dissolved gas, suggesting that removing gas may be a method of increasing the wall heat transfer in low-g. When dissolved gas was present, a much larger primary bubble (most likely composed of gas) was observed, resulting in lower heat transfer and a decrease in the strength of the thermocapillary convection. This directly contradicts the theory proposed by Straub concerning the impact of dissolved gases. Numerical simulations of the flow around the bubbles have also been started, and some results were obtained at low temperature differences. We expect simulations at the temperature differences similar that used in the experiments to be available within the next few months. The test rig was modified so that PIV measurements could be made of the flow around the bubbles in low-g to validate the numerical models. A sheet laser was added to the system, and the fluid was seeded with neutrally buoyant glass spheres. We are currently awaiting a flight opportunity.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
Articles in Peer-reviewed Journals	Henry, C.D. and Kim, J. "Thermocapillary Effects on Low-G Pool Boiling From Microheater Arrays of Various Aspect Ratio" Microgravity Science and Technology, XVI, pp. 170-175 , Jan-2005
Articles in Peer-reviewed Journals	Myers, J.G., Yerramilli, V.K., Hussey, S.W., Yee, G.F., and Kim, J. "Time and space resolved wall temperature and heat flux measurements during nucleate boiling with constant heat flux boundary conditions" ", International Journal of Heat and Mass Transfer, Vol. 48, No. 12, pp. 2429-2442 , Jan-2005
Papers from Meeting Proceedings	Kim, J. "Microgravity and earth gravity boiling heat transfer mechanisms: Investigations using a microheater array" UKC2005, Irvine, CA Aug-2005
Presentation	Henry, C.D., and Kim, J. "Nucleate Pool Boiling Characteristics From a Flat Microheater Array in Low Gravity" Problems of Gas Dynamics and Heat and Mass Transfer in Power Plants", Kaluga, Russia, May 23-27 Aug-2005
Presentation	Kim, J. "Gravity Effects on Boiling Heat Transfer: Investigations Using a Microheater Array" University of Minnesota, Dept. of Mechanical Engineering Seminar Series Feb-2005

Presentation	Kim, J. "Gravity Effects on Boiling Heat Transfer: Investigations Using a Microheater Array" Dept. of Mechanical Engineering, State University of New York at Stony Brook Apr-2005
Presentation	Kim, J. "Phase Change Heat Transfer Mechanisms: Investigations Using a Microheater Array" Intel Corporation Jun-2005